



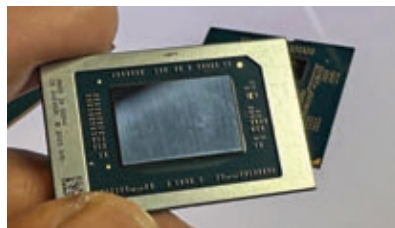
Unleashing AI Power

A closer look at AMD XDNA 2 NPU on the AMD Ryzen AI 300 Series Laptop Processors

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AMD took the wraps off its upcoming AMD Ryzen AI 300 series processors back in June at Computex 2024. We flew down to LA to attend AMD Tech Day 2024, which focused on a deeper dive into the new lineup of AMD's Ryzen AI 300 series processors and beyond. With two days of intense tech and architecture sessions, detailing how AMD has reimagined the upcoming Ryzen AI 300 series of APUs (processors), there is a lot of new and cutting-edge tech in store with the forthcoming AMD Ryzen AI 300 series-powered laptops.

While we will be detailing every aspect of the Zen 5 APU, this article focuses on what is likely to be the most important



aspect of the Zen 5 chip, which is the AMD XDNA 2 NPU, the dedicated neural processing unit that manages all AI tasks. The AMD XDNA 2 NPU boasts up to 50 TOPS of AI performance from the NPU alone.

Ground-up redesign of the NPU

XDNA 2 architecture is based on spatial

data flow architecture; it is a two-dimensional array of AI compute with a flexible interconnect that can be programmed dynamically at runtime and reconfigured to create custom compute memory hierarchies for AI processing. It is a cacheless architecture that promises excellent real-time processing efficiency. The AMD XDNA 2 architecture promises a 5x boost in NPU performance with 32 AI engine tiles (units). AMD is betting big on the XDNA 2 NPU, claiming up to a 5x compute performance boost and a 2x boost in power efficiency. According to AMD's presentation deck, no other NPU hits 50 TOPS of performance (Note: All tests and comparisons are based on AMD's internal testing methodology and process).

The Ryzen AI (XDNA 2) Engine is like a team of specialized workers that can adapt to different tasks. Each worker can adjust their tools (computation and memory resources) to get the job done efficiently, enabling the engine to handle more tasks without using too much energy.

For example, imagine working on a project that requires both writing and graphic design; the XDNA 2 Engine can assign one worker to focus on writing and another to focus on design, using the right tools for each task, making it faster and more efficient. Compared to the previous model (XDNA), the new XDNA 2 NPU can handle five times more tasks while using